

Token export pipeline — one JSON source → every consumable form

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Master technical specification — Volume 10 (Design System), nano-detail deep-dive. This document **owns** the **build pipeline** that turns `helix_design`'s one canonical JSON token source into **every** consumable platform form: the transform tool (a **Style-Dictionary-class** generator — role named, exact tool choice surfaced as a decision), the per-target emit (CSS custom properties `light+dark`, Dart `ThemeExtension + ThemeData`, SwiftUI, Jetpack Compose `Material3 ColorScheme`, ArkTS `HarmonyOS` resources, C/Qt header

- QML singleton), each with a **concrete output sample**, the **drift gate** (generated output **MUST** match a fresh regenerate — a hand-edited derived form fails the build, §11.4.108-style), the §11.4.65 / §11.4.168 export + visual- validation hook, and how OpenDesign's `tokens.css` feeds **in** as one authoring input (D-DESIGN-3 round-trip).

SPEC-ONLY. It describes *the pipeline contract* — the transforms, the output shapes, the gates — not the shipping `helix_design` build. The **token model** (tiers, `$value/$type` envelope, naming, theming axes) is owned by [`design-tokens.md`]; the **colour values + contrast** by [`color-system.md`]; the **type/icon/motion** token families by [`typography-iconography-motion.md`]; the **submodule layout + the export/targets/*.ts placement + OpenDesign relationship** by [`00-overview-and-submodule.md`] (§4, §6, §10) and [`opendesign-foundation.md`]. This document is **original HelixVPN design work** (the pipeline architecture is owned here, layered on the verified OpenDesign facts).

Boundary with sibling docs. Owns: the transform stages, the per-target output samples, the drift gate, the round-trip-with-OpenDesign mechanics, the §11.4.65/.168 export hook. Consumes: the JSON schema + the `$type→target` mapping table [`design-tokens.md` §3.2]; the 8-platform target list + `dist/**-is-a-build-derivative` rule [`00-overview-and-submodule.md` §4/§6, DS-I6]; the verified fact that OpenDesign does **not** emit Dart/Swift/Compose/ ArkTS/C-Qt (so these exporters are `helix_design`'s own) [`opendesign-foundation.md` §0/§1.2].

Evidence base. [DT §N] = final/v10-design/design-tokens.md; [COLOR §N] = final/v10-design/color-system.md; [TIM §N] = final/v10-design/typography-iconography-motion.md; [OV §N] = final/v10-design/00-overview-and-submodule.md; [OD §N] = final/v10-design/opendesign-foundation.md; [SPINE §N] = final/SPECIFICATION.md. Claims not grounded in the evidence base or in this document's own original design choices are tagged UNVERIFIED per constitution §11.4.6 — never fabricated. The **exact** transform-library API shapes (the precise dist/** byte forms) are pinned by the pipeline's own contract test and tagged UNVERIFIED until that test exists.

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0. The pipeline at a glance

helix_design holds **one** authored source of truth — the tiered JSON token tree (tokens/** + presets/**, [DT §3], [OV §4]). The export pipeline (export/) is the machine that turns that one source into the **seven** forms the 3 apps × 8 platforms consume, so **no platform ever re-types a token by hand** (DS-I6, [OV §0.2]). The pipeline is the **enforcement** of the one-source / zero-drift rule: the only way a platform theme can diverge from source is a hand-edit of a generated file, and the drift gate (§5) catches that hand-edit.

#	Stage	Input	Output
1	Load + merge	tokens/ ** JSON files + active presets/ <brand>/	one in-memory token dictionary

#	Stage	Input	Output
2	Resolve references	the dictionary	every {ref} dereferenced to a literal (same-\$type-checked)
3	Theme-split	the resolved dictionary	a light table + a dark table per [DT §5.1]
4	Per-target format	the two theme tables	dist/ {css,dart,swift,compose,arkts,cqt,json}/ **
5	Drift gate	a fresh re-run	byte-compare vs the committed package-source forms (§5)
6	Export + visual-validate	the docs + the emitted forms	§11.4.65 .html/.pdf siblings + §11.4.168 visual validation

flowchart LR

```

    SRC["tokens/** + presets/&lt;brand&gt;/ (authored JSON)"]
    OD["OpenDesign tokens.css (authoring round-trip §7)"]
    OD -.->|"D-DESIGN-3 (b): css→json import"| SRC
    SRC --> L["1 load+merge"] --> R["2 resolve refs"] --> T["3
theme-split (light/dark)"]
    T --> F["4 per-target format"]
    F --> CSS["dist/css/helix.css"]
    F --> DART["dist/dart + ThemeExtension"]
    F --> SWIFT["dist/swift/HelixTokens.swift"]
    F --> COMPOSE["dist/compose/HelixTokens.kt"]
    F --> ARKTS["dist/arkts/helix_tokens.ets"]
    F --> CQT["dist/cqt/helix_tokens.h + Theme.qml"]
    F --> JSON["dist/json/tokens.json"]
    F --> G["5 drift gate (byte-match a fresh re-run)"]
    F --> X["6 §11.4.65 export + §11.4.168 visual-validate"]
    JSON -.->|"export target → tokens.css"| OD

```

1. Inputs — JSON source + the OpenDesign round-trip

1.1 The primary input — the tiered JSON source

The authored input is the three-tier JSON tree [DT §3, OV §4]:

```

helix_design/tokens/
├─ primitive/    color.json type.json space.json radius.json
elevation.json
|               motion.json zindex.json breakpoints.json
icon.json
├─ semantic/    color.light.json color.dark.json type.json
space.json

```

```
|          elevation.json motion.json border.json
└─ component/  connect_button.json status_chip.json
exit_picker.json ...
helix_design/presets/<brand>/  brand.json connection_state.json
```

The active **brand preset** ([OV §5.4]) is selected at build time (`--preset presets/helix/`, default HelixVPN; a future app drops `presets/<x>/`). The preset supplies the values the semantic tier references — the **only** place brand-specific values live (DS-I1, [OV §3]).

1.2 The secondary input — OpenDesign tokens.css (round-trip)

OpenDesign authoring/refinement [OD §6] edits the HelixVPN design system's `tokens.css`. Per **D-DESIGN-3** ([OV §10.2]) the project **leans (b)** — JSON is the source, `tokens.css` is a **generated export target** that OpenDesign consumes/previews; refinement edits made *in* OpenDesign are reconciled **back** into `tokens/**` (a `css→json` import diff). The round-trip touches only the **verified** OpenDesign surface (CSS custom properties, [OD §4]) — never an unverified OpenDesign API. §7 details the mechanics.

2. The transform tool (Style-Dictionary-class) — role & decision

2.1 The role

The transform tool is a **token-build engine** that: loads a tiered token tree, resolves `{ref}` references, applies per-platform **transforms** (e.g. `#RRGGBB` → Dart `Color(0xAARRGGBB)`, `16px` → Compose `16.dp`), and runs per-platform **formatters** that emit a file in the target language. This is exactly the **Style-Dictionary** category of tool (Amazon's open-source design-token build system is the canonical example; the W3C Design-Tokens Community Group format the ecosystem converges on is the `$value/$type` envelope [DT §3.1] this project already authors in).

2.2 The decision

D-EXP-1 (open, §9) — exact transform engine. The pipeline is specified against the **Style-Dictionary-class role** (§2.1), not hard-bound to one library, so the choice is surfaced, not silently made (§11.4.6):

Candidate	Note
Style Dictionary (amzn, v4)	the reference implementation; native W3C-DTCG <code>\$type/\$value</code> support, custom transforms/formats in TS; the lean default
Terrazzo (formerly cobalt-ui)	W3C-DTCG-native, TS-first, plugin per target
a bespoke TS generator in export/build.ts	full control, no dep; more code to own/test

Lean: **Style Dictionary v4** (mature, DTCG-native, custom-format API covers all 6 non-JSON targets), with bespoke formatters for the targets it lacks built-in (ArkTS, C/Qt). The engine is a **build-time dev dependency** declared in `helix-deps.yaml` `build_toolchain` ([OV §7.2]) — never a runtime dependency of the consuming apps. **UNVERIFIED** until the Volume-10 review ratifies the exact library + pins its version per §11.4.99; the *pipeline contract* (stages §3, outputs §4, drift gate §5) holds regardless of which engine fills the role.

Honest boundary (§11.4.6 / [OD §0]). OpenDesign does **NOT** itself emit Dart/Swift/Compose/ArkTS/C-Qt (verified, [OD §1.2]). These exporters are therefore **helix_design's own** Style-Dictionary-class pipeline — *not* an OpenDesign feature. If a future requirement is "OpenDesign itself emits a Compose form", that is an **extend-OpenDesign-upstream** task per §11.4.74 ([OD §7], D-DESIGN-4) — not assumed here.

3. The transform stages (resolve → theme-split → per-target format)

3.1 Stage 1–2 — load, merge, resolve

The loader merges every `tokens/**` file + the active preset into one dictionary, then resolves references depth-first. Resolution is **total** and **type-checked** ([DT §3.1]): a `{ref}` **MUST** resolve to an existing token of the **same \$type**, else the build **fails** (never a silent default, §11.4.6 + [DT §9 CM-token-refs-resolve]).

3.2 Stage 3 — theme-split

Only **themed** categories (color, elevation, a small border/opacity set [DT §3.1/§6]) carry a `{light, dark}` fork. Stage 3 produces **two** fully-resolved flat tables — a `light` table and a `dark` table — where every themed token has collapsed to its theme-specific literal and every non-themed token (space, radius, type metrics, motion, z, breakpoints, [DT §6]) appears identically in both. The two tables are the input to every formatter, so **every** target emits a `light` **and** a `dark` form (a target emitting one theme fails [OV §12 DS-LIGHT-DARK-COMPLETE]).

3.3 Stage 4 — per-target format

Each target is a `transform-set` + a formatter:

Target	Transform highlights (\$type → form, [DT §3.2])	Formatter emits
CSS	<code>color→#RRGGBB[AA]</code> , <code>dim→px/rem, dur→ms</code> , <code>bezier→cubic-bezier()</code>	<code>:root (light) + [data-theme=dark]</code> (dark) custom-property blocks
Dart	<code>color→Color(0xAARRGGBB)</code> , <code>dim→double logical px</code> ,	a <code>tokens</code> file + a <code>ThemeExtension<HelixTokens></code> with <code>.light()/dark()</code>

Target	Transform highlights (\$type → form, [DT §3.2])	Formatter emits
	weight→FontWeight.wNNN, dur→Duration	
SwiftUI	color→Color(.sRGB,...)/ asset-catalog entry, dim→CGFloat, weight→Font.Weight	a HelixTokens.swift + a .xcassets colorset per themed colour
Compose	color→Color(0x...), dim→.dp/.sp, weight→FontWeight.WNNN	a HelixTokens.kt with lightColors/darkColors + an M3 ColorScheme builder
ArkTS	color→\$rrggbb/resource, dim→vp/fp	a helix_tokens.ets + resources/base + resources/ dark qualifier sets
C/Qt	color→QColor("#..."), dim→int px	a helix_tokens.h (#define) + a Theme.qml singleton (light/dark property sets)
JSON	identity (resolved)	dist/json/tokens.json — the flattened resolved source (other tooling + the §7 css round-trip read it)

4. Per-target output samples (concrete)

Every sample below is a **valid, self-consistent** emit of the **same** small token slice — `color.semantic.surface.default, text.primary, action.primary, state.connected.fill, state.danger.fill, font.semantic.title.lg, space.scale.4, radius.scale.full, motion.semantic.connectPulse` — in **both** themes. Colour literals match [COLOR §2/§3]; type metrics match [TIM §1.1]; scales match [DT §6]. **No placeholders, no TODO** (§11.4.6).

4.1 CSS custom properties (light + dark)

`dist/css/helix.css` — light on `:root`, dark on `[data-theme="dark"]` (consumed by the Console Web build + any web chrome, [OV §6]). A `@media (prefers-color-scheme: dark)` block mirrors the attribute so OS dark works without an explicit `data-theme`.

```
/* dist/css/helix.css – GENERATED, do not edit (drift gate §5) */
:root {
  /* color – light */
  --hx-color-surface-default: #FFFFFF;
  --hx-color-text-primary: #151920;
  --hx-color-action-primary: #2F47C4;
  --hx-color-state-connected-fill: #15803D;
  --hx-color-state-danger-fill: #DC2626;
  /* type (theme-invariant) */
  --hx-font-title-lg-family: 'Inter', system-ui, sans-serif;
```

```

--hx-font-title-lg-size: 20px;
--hx-font-title-lg-line: 1.30;
--hx-font-title-lg-weight: 600;
/* scales (theme-invariant) */
--hx-space-4: 16px;
--hx-radius-full: 9999px;
--hx-motion-connect-pulse-dur: 1200ms;
--hx-motion-connect-pulse-ease: cubic-bezier(0.4, 0, 0.6, 1);
}
[data-theme="dark"] {
  --hx-color-surface-default: #151920;
  --hx-color-text-primary: #EDEFF3;
  --hx-color-action-primary: #3D5AF1;
  --hx-color-state-connected-fill: #22C55E;
  --hx-color-state-danger-fill: #F87171;
}
@media (prefers-color-scheme: dark) {
  :root:not([data-theme="light"]) {
    --hx-color-surface-default: #151920;
    --hx-color-text-primary: #EDEFF3;
    --hx-color-action-primary: #3D5AF1;
    --hx-color-state-connected-fill: #22C55E;
    --hx-color-state-danger-fill: #F87171;
  }
}
@media (prefers-reduced-motion: reduce) {
  :root { --hx-motion-connect-pulse-dur: 0ms; } /* [TIM §8.1] */
}

```

4.2 Dart — ThemeExtension + ThemeData

dist/dart/helix_tokens.dart — a ThemeExtension carrying every token so a widget reads Theme.of(context).extension<HelixTokens>()!; plus a ThemeData light/dark pair the helix-ui MaterialApp(theme:, darkTheme:) consumes ([DT §5.1], [OV §6]).

```

// dist/dart/helix_tokens.dart – GENERATED, do not edit (drift
// gate §5)
import 'package:flutter/material.dart';

@immutable
class HelixTokens extends ThemeExtension<HelixTokens> {
  final Color surfaceDefault, textPrimary, actionPrimary,
    stateConnected, stateDanger;
  final Duration connectPulseDur;
  const HelixTokens({
    required this.surfaceDefault, required this.textPrimary,
    required this.actionPrimary,
    required this.stateConnected, required this.stateDanger,
    required this.connectPulseDur,
  });

  static const light = HelixTokens(

```

```

        surfaceDefault: Color(0xFFFFFFFF), textPrimary:
            Color(0xFF151920),
        actionPrimary: Color(0xFF2F47C4), stateConnected:
            Color(0xFF15803D),
        stateDanger: Color(0xFFDC2626), connectPulseDur:
            Duration(milliseconds: 1200));
static const dark = HelixTokens(
    surfaceDefault: Color(0xFF151920), textPrimary:
        Color(0xFFEDEFF3),
    actionPrimary: Color(0xFF3D5AF1), stateConnected:
        Color(0xFF22C55E),
    stateDanger: Color(0xFFF87171), connectPulseDur:
        Duration(milliseconds: 1200));

```

@override

```

HelixTokens copyWith({Color? surfaceDefault, Color?
    textPrimary, Color? actionPrimary,
    Color? stateConnected, Color? stateDanger, Duration?
    connectPulseDur}) => HelixTokens(
    surfaceDefault: surfaceDefault ?? this.surfaceDefault,
    textPrimary: textPrimary ?? this.textPrimary,
    actionPrimary: actionPrimary ?? this.actionPrimary,
    stateConnected: stateConnected ?? this.stateConnected,
    stateDanger: stateDanger ?? this.stateDanger,
    connectPulseDur: connectPulseDur ?? this.connectPulseDur);

```

@override

```

HelixTokens lerp(ThemeExtension<HelixTokens>? other, double t) {
    if (other is! HelixTokens) return this;
    return HelixTokens(
        surfaceDefault: Color.lerp(surfaceDefault,
            other.surfaceDefault, t)!,
        textPrimary: Color.lerp(textPrimary, other.textPrimary, t)!,
        actionPrimary: Color.lerp(actionPrimary,
            other.actionPrimary, t)!,
        stateConnected: Color.lerp(stateConnected,
            other.stateConnected, t)!,
        stateDanger: Color.lerp(stateDanger, other.stateDanger, t)!,
        connectPulseDur: t < 0.5 ? connectPulseDur :
            other.connectPulseDur);
    }
}

```

```

const _titleLg = TextStyle(fontFamily: 'Inter', fontSize: 20,
    height: 1.30, fontWeight: FontWeight.w600);

```

```

ThemeData helixLight() => ThemeData(
    useMaterial3: true, brightness: Brightness.light,
    scaffoldBackgroundColor: const Color(0xFFFFFFFF),
    colorScheme: const ColorScheme.light(
        primary: Color(0xFF2F47C4), surface: Color(0xFFFFFFFF),
        onSurface: Color(0xFF151920)),
    textTheme: const TextTheme(titleLarge: _titleLg),
    extensions: const [HelixTokens.light]);

```



```
ThemeData helixDark() => ThemeData(
  useMaterial3: true, brightness: Brightness.dark,
  scaffoldBackgroundColor: const Color(0xFF151920),
  colorScheme: const ColorScheme.dark(
    primary: Color(0xFF3D5AF1), surface: Color(0xFF151920),
    onSurface: Color(0xFFEDEF3)),
  textTheme: const TextTheme(titleLarge: _titleLg),
  extensions: const [HelixTokens.dark]);
```

Dart colour is Color(0xAARRGGBB) — the leading FF is full alpha; an 8-hex token like #1519201A ([DT §4] light-card shadow) emits Color(0x1A151920).

4.3 SwiftUI — generated tokens + asset catalog

dist/swift/HelixTokens.swift — a generated Swift file referencing **asset-catalog colorsets** (the idiomatic SwiftUI way to carry light+dark: one named colorset holds both appearances, resolved by @Environment(\.colorScheme) automatically). The exporter emits both the .swift accessor and the .xcassets colorset JSON.

```
// dist/swift/HelixTokens.swift – GENERATED, do not edit (drift
// gate §5)
import SwiftUI

public enum HelixTokens {
  // colours resolve light/dark automatically via the asset
  // catalog (below)
  public static let surfaceDefault = Color("HXSurfaceDefault",
    bundle: .module)
  public static let textPrimary = Color("HXTextPrimary",
    bundle: .module)
  public static let actionPrimary = Color("HXActionPrimary",
    bundle: .module)
  public static let stateConnected = Color("HXStateConnected",
    bundle: .module)
  public static let stateDanger = Color("HXStateDanger",
    bundle: .module)
  // type (theme-invariant)
  public static let titleLg = Font.custom("Inter", size:
    20).weight(.semibold)
  // scale / motion (theme-invariant)
  public static let space4: CGFloat = 16
  public static let radiusFull: CGFloat = 9999
  public static let connectPulse: Double = 1.200 // seconds
}

// dist/swift/Resources/HelixColors.xcassets/
// HXStateConnected.colorset/Contents.json
// GENERATED – one colorset per themed colour, light + dark
// appearance
{
  "colors": [
    { "idiom": "universal",
```

```

        "color": { "color-space": "srgb",
            "components": { "red": "0x15", "green": "0x80", "blue":
                "0x3D", "alpha": "1.000" } } },
    { "idiom": "universal",
        "appearances": [ { "appearance": "luminosity", "value":
            "dark" } ],
        "color": { "color-space": "srgb",
            "components": { "red": "0x22", "green": "0xC5", "blue":
                "0x5E", "alpha": "1.000" } } }
    ],
    "info": { "author": "helix_design exporter", "version": 1 }
}

```

The native shim's font is the **system font** for the tiny NE-extension surface ([TIM §2.2 D-TYPE-1]) — `Font.custom("Inter",...)` above is for any iOS/macOS surface that *does* bundle the brand font; the colorset (palette) reaches **all** native surfaces regardless of font.

4.4 Jetpack Compose — Material3 ColorScheme + Typography

`dist/compose/HelixTokens.kt` — a Compose object plus `lightColorScheme()` / `darkColorScheme()` Material3 builders the Android quick-settings tile / notification surfaces consume ([OV §6]).

```

// dist/compose/HelixTokens.kt – GENERATED, do not edit (drift
// gate §5)
package com.helixvpn.design

import androidx.compose.material3.*
import androidx.compose.ui.graphics.Color
import androidx.compose.ui.text.TextStyle
import androidx.compose.ui.text.font.FontWeight
import androidx.compose.ui.unit.dp
import androidx.compose.ui.unit.sp

object HelixColors {
    // raw tokens (both themes)
    val SurfaceLight = Color(0xFFFFFFFF); val SurfaceDark =
        Color(0xFF151920)
    val OnSurfaceLight = Color(0xFF151920); val OnSurfaceDark =
        Color(0xFFEDEF3)
    val ActionLight = Color(0xFF2F47C4); val ActionDark =
        Color(0xFF3D5AF1)
    val ConnectedLight = Color(0xFF15803D); val ConnectedDark =
        Color(0xFF22C55E)
    val DangerLight = Color(0xFFDC2626); val DangerDark =
        Color(0xFFFF87171)
}

val HelixLightColorScheme: ColorScheme = lightColorScheme(
    primary = HelixColors.ActionLight, surface =
        HelixColors.SurfaceLight,
    onSurface = HelixColors.OnSurfaceLight, error =
        HelixColors.DangerLight)
val HelixDarkColorScheme: ColorScheme = darkColorScheme(

```

```

primary = HelixColors.ActionDark, surface =
    HelixColors.SurfaceDark,
onSurface = HelixColors.OnSurfaceDark, error =
    HelixColors.DangerDark)

object HelixType {
    val TitleLg = TextStyle(fontSize = 20.sp, lineHeight = 26.sp,
        fontWeight = FontWeight.W600)
}
object HelixScale { val Space4 = 16.dp; val RadiusFull = 9999.dp }
object HelixMotion { const val ConnectPulseMs = 1200L }
// "connected" is not an M3 slot → exposed as a HelixColors token
the StatusChip reads.

```

Material3's ColorScheme has no "connected/danger-state" slot, so the connection-state colours ([COLOR §3]) are exposed as **HelixColors tokens** the StatusChip/ConnectButton read directly (the [DT §8] component-token seam), while the M3 slots (primary/surface/error) carry the brand + error mapping for stock M3 widgets.

4.5 ArkTS — HarmonyOS resources / AppStorage

dist/arkts/helix_tokens.ets + the HarmonyOS resource qualifier tree — light in resources/base/element/, dark in resources/dark/element/ (HarmonyOS resolves the qualifier by system theme), consumed by the HarmonyOS VPN-ability surfaces ([OV §6]).

```

// dist/arkts/helix_tokens.ets – GENERATED, do not edit (drift
gate §5)
export class HelixTokens {
    // colour tokens resolve light/dark via the resource qualifier
    tree (below)
    static readonly surfaceDefault: Resource =
        $r('app.color.hx_surface_default');
    static readonly textPrimary: Resource =
        $r('app.color.hx_text_primary');
    static readonly actionPrimary: Resource =
        $r('app.color.hx_action_primary');
    static readonly stateConnected: Resource =
        $r('app.color.hx_state_connected');
    static readonly stateDanger: Resource =
        $r('app.color.hx_state_danger');
    // type / scale / motion (theme-invariant)
    static readonly titleLgSize: number = 20;           // fp
    static readonly titleLgWeight: number = 600;
    static readonly space4: number = 16;                // vp
    static readonly radiusFull: number = 9999;          // vp
    static readonly connectPulseMs: number = 1200;
}

// dist/arkts/resources/base/element/color.json (LIGHT – system
default)
{ "color": [
    { "name": "hx_surface_default", "value": "#FFFFFF" },

```

```

    { "name": "hx_text_primary",    "value": "#151920" },
    { "name": "hx_action_primary",  "value": "#2F47C4" },
    { "name": "hx_state_connected", "value": "#15803D" },
    { "name": "hx_state_danger",    "value": "#DC2626" }
  ] }
// dist/arkts/resources/dark/element/color.json (DARK – resolved
// when system dark)
{ "color": [
  { "name": "hx_surface_default", "value": "#151920" },
  { "name": "hx_text_primary",    "value": "#EDEFF3" },
  { "name": "hx_action_primary",  "value": "#3D5AF1" },
  { "name": "hx_state_connected", "value": "#22C55E" },
  { "name": "hx_state_danger",    "value": "#F87171" }
] }

```

UNVERIFIED (U-EXP-1). The exact ArkTS `$r('app.color....')` resolver key form + the `resources/dark/` qualifier directory name are HarmonyOS-SDK-version dependent; stated as the *intended* mapping per [DT §3.2 U-DT-1], pinned + re-verified per §11.4.99 against the live HarmonyOS docs before the ArkTS exporter ships. Not asserted as fact (§11.4.6).

4.6 C / Qt — header + QML singleton

`dist/cqt/helix_tokens.h` (#define constants for C/C++) + `dist/cqt/Theme.qml` (a QML singleton exposing a light/dark property set, switched by a theme property) — consumed by the Aurora Qt/C++ surfaces ([OV §6]).

```

// dist/cqt/helix_tokens.h – GENERATED, do not edit (drift gate
// §5)
#ifndef HELIX_TOKENS_H
#define HELIX_TOKENS_H
/* light */
#define HX_LIGHT_SURFACE_DEFAULT "#FFFFFF"
#define HX_LIGHT_TEXT_PRIMARY    "#151920"
#define HX_LIGHT_ACTION_PRIMARY  "#2F47C4"
#define HX_LIGHT_STATE_CONNECTED "#15803D"
#define HX_LIGHT_STATE_DANGER    "#DC2626"
/* dark */
#define HX_DARK_SURFACE_DEFAULT  "#151920"
#define HX_DARK_TEXT_PRIMARY     "#EDEFF3"
#define HX_DARK_ACTION_PRIMARY   "#3D5AF1"
#define HX_DARK_STATE_CONNECTED  "#22C55E"
#define HX_DARK_STATE_DANGER     "#F87171"
/* type / scale / motion (theme-invariant) */
#define HX_FONT_TITLE_LG_SIZE_PX 20
#define HX_FONT_TITLE_LG_WEIGHT  600
#define HX_SPACE_4_PX             16
#define HX_RADIUS_FULL_PX         9999
#define HX_MOTION_CONNECT_PULSE_MS 1200
#endif /* HELIX_TOKENS_H */

```

```
// dist/cqt/Theme.qml – GENERATED QML singleton, do not edit
(drift gate §5)
pragma Singleton
import QtQuick 2.15
QtObject {
    property bool dark: false
    readonly property color surfaceDefault: dark ? "#151920" :
        "#FFFFFF"
    readonly property color textPrimary:    dark ? "#EDEFF3" :
        "#151920"
    readonly property color actionPrimary:  dark ? "#3D5AF1" :
        "#2F47C4"
    readonly property color stateConnected: dark ? "#22C55E" :
        "#15803D"
    readonly property color stateDanger:    dark ? "#F87171" :
        "#DC2626"
    readonly property int   titleLgSize:    20
    readonly property int   titleLgWeight: 600
    readonly property int   space4:         16
    readonly property int   radiusFull:     9999
    readonly property int   connectPulseMs: 1200
}
```

UNVERIFIED (U-EXP-2). The exact QML singleton registration form (qml dir singleton Theme 1.0 Theme.qml) and whether Aurora's Qt build uses a resource (.qrc) vs. a module import is Aurora-Qt-version dependent; pinned + re-verified per §11.4.99 before the C/Qt exporter ships ([DT §3.2 U-DT-1]).

5. The drift gate (generated MUST match source)

The drift gate is the **load-bearing** §11.4.108-class mechanism: it makes a hand-edit of a generated form a **build failure**, so the only way to change a platform's tokens is to change the JSON source + regenerate.

5.1 What it asserts

DS-DRIFT (the §11.4.108 SOURCE→ARTIFACT equality gate for tokens):

1. run `pnpm export` into a CLEAN temp dir
2. byte-compare each freshly-generated form against the committed package-source form
3. ANY difference ⇒ FAIL (the committed form was hand-edited or the source drifted)

The comparison is **byte-for-byte** (after a deterministic format pass, so formatter non-determinism — map ordering, trailing newline — is normalised at generation, never papered over at compare time). A generated file carries a **GENERATED, do not edit** banner + a content hash header; the gate also asserts the hash matches the file body (a hand-edit that forgets to update the banner is still caught by the byte-compare).

5.2 Why byte-compare, not "looks equivalent"

Per §11.4.108 (runtime-signature / artifact-byte equality) and §11.4.6 (no-guessing): "the Dart and the CSS *look* like the same colour" is not a proof the source and the artifact agree. The byte-compare against a fresh re-run is the **captured evidence** that `dist/**` (and the checked-in package-source Dart, [OV §4 D-DESIGN-2]) is exactly what the current `tokens/**` produces — no drift, no stale form from a previous source state.

5.3 The paired mutation (§1.1)

Gate	Mutation that MUST make it FAIL
DS-DRIFT	hand-edit one hex in <code>helix_tokens.dart</code> (e.g. <code>0xFF2F47C4</code> → <code>0xFF000000</code>) without touching <code>tokens/**</code> → the fresh re-run differs → FAIL
DS-DRIFT (source side)	change a <code>tokens/**</code> value but commit the old <code>dist/**</code> → fresh re-run differs → FAIL (forces regenerate-on-source-change)
DS-EXPORT-FORM-VALID ([OV §12])	emit syntactically-broken Dart (drop a <code>;</code>) → <code>dart analyze</code> fails → FAIL (the form must also <i>parse</i> in its toolchain, §11.4.110 build-readiness)
DS-LIGHT-DARK-COMPLETE ([OV §12])	delete the dark colorset appearance / the <code>[data-theme=dark]</code> block → a theme is missing → FAIL ([TIM]/[COLOR] light+dark mandate)

5.4 Where the gate runs

flowchart LR

```
edit["edit tokens/** OR (forbidden) edit a dist/** form"]
edit --> pc["pre-commit hook (§11.4.75)"]
pc -->|"runs DS-DRIFT on staged"| v{match fresh re-run?}
v -->|"no"| block["BLOCK commit – regenerate or revert the hand-edit"]
v -->|"yes"| ci["helix_design CI / pre-build (§11.4.32)"]
ci -->|"DS-DRIFT + DS-EXPORT-FORM-VALID + DS-LIGHT-DARK-COMPLETE"| go["GO"]
consumer["consuming app pre-build (helix-ui)"] -->|"pointer bump bumps tokens → re-run drift + visual-regression (DS-I5)"| go
```

The gate runs at three seams: the **pre-commit** hook (§11.4.75, blocks a hand-edit landing), `helix_design CI/pre-build` (§11.4.32 post-pull sweep), and the **consuming app's** pre-build when a submodule-pointer bump changes `tokens` ([OV §8] — a token change triggers the app's visual-regression suite, DS-I5).

6. The §11.4.65 / §11.4.168 export + visual-validation hook

6.1 §11.4.65 — Markdown export of the pipeline + token docs

Every Volume-10 doc (this one included) travels in the submodule (`helix_design/docs/v10-design/`, [OV §4]) and is a §11.4.65-scope Markdown doc: each edit regenerates synchronized `.html` + `.pdf` siblings, and (for the §11.4.153 feature-status class) `.docx`. The pipeline's own run produces a machine-readable **export manifest** (`dist/.export-manifest.json` — every emitted form + its content hash + the source-commit it was generated from) that the docs reference, so a reader of the rendered token reference sees exactly which source produced it.

6.2 §11.4.168 — visual validation of the rendered token reference

Because the token reference docs contain **rendered swatches** + **Mermaid diagrams** (this doc's pipeline diagram, the [COLOR] swatch map, the [TIM] type-scale diagram), the §11.4.168 exported-document validation applies: an **independent** check confirms the exported `.pdf/.html` (a) carries the content faithfully, (b) shows **no raw Mermaid/diagram source as body text** (the §11.4.168 forensic failure), and (c) renders every diagram as an **image** + every swatch as a visible colour block. `pdftotext` catches raw source leaking as text; `pdfimages` confirms the diagrams rasterised; the self-validated golden-good/golden-bad analyzer ([§11.4.107(10)]) is seeded with a known-bad export (raw flowchart source in the body) that **MUST FAIL** the gate.

6.3 The combined hook

```
flowchart LR
  src["edit tokens/** OR a v10-design doc"]
  src --> exp["pnpm export (forms) + sync_all_markdown_exports (docs)"]
  exp --> drift["DS-DRIFT (§5)"]
  exp --> mdsync["§11.4.65: .html/.pdf(/.docx) siblings refreshed"]
  mdsync --> vv["§11.4.168 visual validation:<br/>pdftotext (no raw diagram source) +<br/>pdfimages (diagrams rasterised) +<br/>swatch-block present"]
  drift --> gate{all G0?}
  vv --> gate
  gate -->|no| iterate["fix → re-export → re-validate (§11.4.134 iterate-to-G0)"]
  gate -->|yes| done["commit / publish"]
```

A token-doc PDF that ships raw flowchart TD ... as readable text, or a swatch table with no rendered colour blocks, is a §11.4.168 violation — a release blocker — caught here, never shipped (the [COLOR §5]/[TIM §4] no-overlay rule extended to the *documentation* of the design system).

7. How OpenDesign's tokens.css feeds in

7.1 The round-trip (D-DESIGN-3, lean (b))

Per [OV §10.2] / [OD §4–§5] the project leans **JSON-as-source**: `tokens.css` is a **generated export target** (one more formatter, `export/targets/css.ts` → `opendesign/helix/tokens.css`) that OpenDesign consumes as the active design system; refinement edits a designer makes *in* OpenDesign land in that `tokens.css`, and a **css→json reconciliation** imports the diff back into `tokens/**` so the polyglot exports stay in sync (DS-I6).

sequenceDiagram

```
participant J as tokens/** (JSON source of truth)
participant E as export/targets/css.ts
participant ODC as opendesign/helix/tokens.css
participant OD as OpenDesign (desktop / od CLI / MCP)
participant REC as css→json reconciler
J->>E: pnpm export (themed, light+dark)
E->>ODC: write tokens.css (:root + [data-theme=dark])
ODC->>OD: set active system; designer/agent refines
OD-->>ODC: refinement edits (tokens.css tweaks)
ODC->>REC: diff tokens.css vs last-generated
REC->>J: apply the human edits back into tokens/** (same-
```

\$type)

```
J->>E: re-export ALL polyglot forms (drift gate §5 + visual §6)
```

7.2 Why the round-trip is safe (only verified OpenDesign surface)

The round-trip touches **only** `tokens.css` — CSS custom properties, the **verified** OpenDesign token form ([OD §3.3/§4]). It does **not** depend on any **UNVERIFIED** OpenDesign API (no theme-engine knobs, no multi-language export OD lacks, [OD §1.2]). The `css→json` reconciler is `helix_design`-owned (a CSS parser → token-path mapper); it maps `--hx-color-state-connected-fill` [?] `color.semantic.state.connected.fill` by the [DT §2] naming convention (`--hx-` prefix + `-` for `.`). A `tokens.css` edit that does not map cleanly to a known token path is surfaced to the operator (§11.4.6 — never silently dropped or guessed), not auto-merged.

UNVERIFIED (U-EXP-3). Whether OpenDesign's renderer honours the `[data-theme="dark"]` block the css exporter emits, or expects a different dark-mode convention (twin "HelixVPN Light"/"HelixVPN Dark" systems, `prefers-color-scheme`) is [OD §5.2 U4] — confirmed against a live built-in OpenDesign `tokens.css` in the refinement pass. If twin systems are required, the css exporter emits `opendesign/helix-light/tokens.css` + `opendesign/helix-dark/tokens.css` from the same JSON source — a **formatter option**, not a token-source change (the JSON stays the single source). Not asserted as fact (§11.4.6).

8. The full pipeline diagram

flowchart TB

```
    subgraph IN["INPUTS"]
        JSON["tokens/** + presets/<brand>/ (authored JSON,
tiered, light+dark)"]
        ODCSS["OpenDesign tokens.css (round-trip $7)"]
    end
    ODCSS -.→| "css→json reconcile (D-DESIGN-3 b)" | JSON
    JSON --> S1["1 load + merge (active preset)"]
    S1 --> S2["2 resolve {refs} (same-$type, total - $3.1)"]
    S2 --> S3["3 theme-split → light table + dark table ($3.2)"]
    S3 --> S4["4 per-target format (Style-Dictionary-class - $2,
$3.3)"]

    S4 --> O1["dist/css/helix.css (:root + [data-theme=dark] +
@media - $4.1)"]
    S4 --> O2["dist/dart + ThemeExtension + helixLight()/
helixDark() ($4.2)"]
    S4 --> O3["dist/swift/HelixTokens.swift + .xcassets colorsets
($4.3)"]
    S4 --> O4["dist/compose/HelixTokens.kt + light/darkColorScheme
($4.4)"]
    S4 --> O5["dist/arkts/helix_tokens.ets + resources/{base,dark}
($4.5)"]
    S4 --> O6["dist/cqt/helix_tokens.h + Theme.qml singleton
($4.6)"]
    S4 --> O7["dist/json/tokens.json (resolved) → feeds tokens.css
($7)"]
    O7 -.→ ODCSS

    O1 & O2 & O3 & O4 & O5 & O6 & O7 --> G5["5 DS-DRIFT byte-gate
+ DS-EXPORT-FORM-VALID + DS-LIGHT-DARK-COMPLETE ($5)"]
    G5 --> G6["6 §11.4.65 export + §11.4.168 visual-validate
($6)"]
    G6 --> APPS

    subgraph APPS["CONSUMING APPS (3 apps × 8 platforms)"]
        FL["Flutter helix-ui → Client / Console / Connector
(Dart)"]
        WEB["Console Web (helix.css)"]
        IOS["iOS/macOS native shim (Swift colorsets)"]
        AND["Android tile/notif (Compose)"]
        HM["HarmonyOS ability (ArkTS)"]
        AU["Aurora Qt surface (C/Qt)"]
    end
    O2 --> FL
    O1 --> WEB
    O3 --> IOS
    O4 --> AND
    O5 --> HM
    O6 --> AU
```

9. Surfaced decisions & cross-doc contracts

id	Decision / contract	Status
D-EXP-1 UNVERIFIED	Transform engine = a Style-Dictionary-class tool; lean Style Dictionary v4 (DTCG-native) + bespoke formatters for ArkTS/C-Qt. Build-time dev dep only. Exact library + pinned version ratified at Volume-10 review per §11.4.99.	open
D-EXP-2	Theme is handled by theme-split (Stage 3) into two resolved tables, so every target emits light and dark; no target emits one theme ([OV §12 DS-LIGHT-DARK-COMPLETE]).	decided
D-EXP-3	<code>tokens.css</code> is a generated export target (D-DESIGN-3 (b)), OpenDesign consumes it; refinement round-trips back via the <code>css→json</code> reconciler (§7) — JSON stays the single source.	recommended (leans (b))
D-EXP-4	SwiftUI light/dark via asset-catalog colorsets (idiomatic), not runtime <code>colorScheme</code> branching; Compose via <code>light/darkColorScheme</code> ; ArkTS via <code>resources/{base,dark}</code> ; CSS via <code>[data-theme]</code> + <code>@media</code> ; C/Qt via a dark-property QML singleton.	decided (per §4)
C-EXP-A (consumes)	The JSON schema + the <code>\$type→target</code> mapping table [DT §3.2] is this pipeline's input contract ; a <code>\$type</code> added there is a new transform here.	contract
C-EXP-B (consumes)	The 8-platform target list + the <code>dist/**-is-a-build-derivative</code> + <code>checked-in-package-Dart</code> rule [OV §4/§6, D-DESIGN-2].	contract
C-EXP-C (consumes)	The verified fact OpenDesign does not emit Dart/Swift/Compose/ArkTS/C-Qt [OD §1.2] — so these exporters are <code>helix_design</code> -owned, not an OpenDesign feature.	contract
C-EXP-D (provides)	The drift gate (§5) is the §11.4.108 <code>SOURCE→ARTIFACT</code> equality proof the whole design system rests on; the §6 visual-validation is the §11.4.168 proof for the rendered token docs.	contract
U-EXP-1 UNVERIFIED	Exact ArkTS <code>\$r('app.color...')</code> resolver + <code>resources/dark/</code> qualifier form (HarmonyOS-SDK-version dependent) — pinned + re-verified per §11.4.99 before the ArkTS exporter ships.	open
U-EXP-2 UNVERIFIED	Exact QML singleton registration (<code>qmlDir/.qrc</code>) + Aurora-Qt module form — pinned + re-verified per §11.4.99 before the C/Qt exporter ships.	open

id	Decision / contract	Status
U-EXP-3 UNVERIFIED	OpenDesign's canonical dark-mode tokens.css convention ([data-theme] vs prefers-color-scheme vs twin systems, [OD §5.2 U4]) — confirmed against a live built-in system in the refinement pass; css exporter has a twin-systems formatter option ready.	open

Sources verified

- **The pipeline architecture (stages, theme-split, per-target transforms + formatters, the drift gate, the round-trip mechanics, the §11.4.65/.168 export hook) and every per-target output sample (§4) — NO external source needed — original HelixVPN design work**, layered on the well-established Style-Dictionary-class token-build pattern and each platform's idiomatic theme carrier (CSS custom properties + [data-theme]; Flutter ThemeExtension/ ThemeData; SwiftUI asset-catalog colorsets; Compose Material3 ColorScheme; HarmonyOS resources/{base,dark} qualifiers; Qt QML singleton). The exact per-platform idioms are the documented conventions of each framework; the *pipeline* that emits them is HelixVPN's own design.
- **The W3C Design-Tokens Community-Group \$value/\$type envelope + the Style-Dictionary build-engine category (Stage transforms/formatters) — the DTCG format the token source [DT §3.1] already authors in, and the Style-Dictionary (amzn) reference implementation of a token-build engine; the exact engine + version is D-EXP-1 UNVERIFIED, ratified + pinned per §11.4.99 at the Volume-10 review (not asserted as a settled fact here).**
- **The token model (tiers, \$value/\$type envelope, naming, theming axes, the \$type→target mapping), the colour hexes, the type metrics, the scales — final/v10-design/design-tokens.md §3/§5/§6, final/v10-design/color-system.md §2/§3, final/v10-design/typography-iconography-motion.md §1/§6 (siblings, this wave, read 2026-06-25). Every literal in the §4 samples is reproduced from those documents to keep the samples valid (no placeholders).**
- **The submodule layout (export/targets/*.ts, dist/**-is-a-build-derivative, the checked-in package-Dart D-DESIGN-2), the 8-platform target list, the OpenDesign relationship + the verified absence of OpenDesign multi-language export — final/v10-design/00-overview-and-submodule.md §4/§6/§10 + final/v10-design/opendesign-foundation.md §0/§1.2/§3.3/§4/§5.2 (siblings, read 2026-06-25). The "OpenDesign does not emit Dart/Swift/Compose/ArkTS/C-Qt" finding is grounded there (web-verified 2026-06-25), not assumed.**
- **Constitution clauses §11.4.6 (no-guessing), §11.4.32 (post-pull validation sweep), §11.4.65 (universal Markdown export), §11.4.75 (mechanical enforcement / pre-commit), §11.4.99 (latest-source cross-reference), §11.4.107(10) (self-validated analyzer), §11.4.108 (SOURCE→ARTIFACT equality — the drift gate's class), §11.4.110 (build-readiness / form-must-parse), §11.4.134 (iterate-to-GO), §11.4.162 (OpenDesign mandate), §11.4.168 (exported-doc visual validation) — constitution submodule text embedded in this repo's constitution/CLAUDE.md, accessed 2026-06-25.**
- Items explicitly marked UNVERIFIED (U-EXP-1 ArkTS resolver, U-EXP-2 QML registration, U-EXP-3 OpenDesign dark-mode convention, D-EXP-1

exact engine) are pending their named contract test / §11.4.99 verification pass per §11.4.6 — not asserted as fact.